数学物理方程期中试题(2022-04-20)

姓名: 学号:

共2页，6道题.

1. (10分) 我们已经熟悉均匀细杆的纵振动方程: $\frac{\partial^2 u}{\partial t^2} = \frac{E}{\rho}\frac{\partial^2 u}{\partial x^2}$, 其中常数$E$为杆的杨氏模量，$\rho$为密度. 若细杆并不均匀，其材料成分随$x$周期变化, 导致密度和杨氏模量相应变化: $\rho(x) = \rho_0 + \rho_1\cos(\frac{2\pi x}{l})$, $E(x) = E_0 + E_1\cos(\frac{2\pi x}{l})$, 其中$\rho_0, \rho_1, E_0, E_1, l$ 为常数，写下此杆的纵振动方程. (不要求写推导过程)

2. (10分) 考虑一根端点位于$x = 0$的半无限长均匀柔软弦, 初始时刻弦处于平衡位置。若以一定频率横向摇动端点，会激起横向振动沿弦传播，其服从的方程为:

$$\begin{aligned}
&\frac{\partial^2 u}{\partial t^2} = a^2\frac{\partial^2 u}{\partial x^2}, \quad x > 0, \quad t > 0,\\
&u|_{t=0} = 0, \quad \left.\frac{\partial u}{\partial t}\right|_{t=0} = 0, \quad x > 0,\\
&u|_{x=0} = A\sin(\omega t).
\end{aligned}$$

其中A, ω均为大于零的常数. 写下$u(x,t)$ $(x > 0, t > 0)$ 的表达式(不要求写推导过程)，并画出$t_1 = \frac{\pi}{2\omega}$和$t_2 = \frac{\pi}{\omega}$ 这两个时刻u的波形草图.

3. 方程标准形与求解问题(试题末尾附录中的公式可能有用).

1) (12分) 求合适的新变量，将如下方程化为标准形式并指出方程类型:

$$\frac{\partial^2 u}{\partial x^2} + 2\cos x\frac{\partial^2 u}{\partial x\partial y} - \sin^2 x\frac{\partial^2 u}{\partial y^2} - \sin x\frac{\partial u}{\partial y} = 0.$$

2) (8分) 首先，写出方程通解, 即包含两个任意函数(不是“任意常数”)的解. 然后，考虑区域$-\infty < x < \infty, y > \sin x$，给定其边界条件:

$$u|_{y=\sin x} = \phi(x), \quad \left.\frac{\partial u}{\partial y}\right|_{y=\sin x} = \psi(x),$$

求方程在区域$-\infty < x < \infty, y > \sin x$上的解.

4. (20分) 一个“无限深盒”四壁电势为0，盒底电势为常数V, 求盒中电势$u(x,y,z)$，即求解拉普拉斯方程：

$$\begin{aligned}
&\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0, \qquad 0 < x < a, \quad 0 < y < b, \quad 0 < z < +\infty,\\
&u|_{x=0} = u|_{x=a} = u|_{y=0} = u|_{y=b} = 0,\\
&u|_{z=0} = V, \quad u|_{z=+\infty} = 0,
\end{aligned}$$

其中V为常数.

5. (20分) 有一根长为l, 侧面绝热，而初始时刻$t = 0$时温度为0的均匀细杆，其一端$x = l$ 处温度保持为0，而另一端$x = 0$处温度随时间直线上升，即$u|_{x=0} = At$ (A 为常数). 由于杆很细，其任意横截面上的温度只依赖于x和t，不依赖于横向坐标，因此温度分布$u(x,t)$服从一维热传导方程$\frac{\partial u}{\partial t} = \kappa\frac{\partial^2 u}{\partial x^2}$ $(0 < x < l,$ $t > 0)$. 求此温度分布.

6. 梁的横振动问题.

1) (4分) 写出如下四阶常微分方程的通解(通解包含四个任意常数).

$$X^{(4)}(x) - \lambda X(x) = 0,$$

其中$\lambda > 0$, $X^{(4)}(x)$ 表示$X(x)$的四阶导数.

2) (8分) 求如下问题的本征值与本征函数:

$$\begin{aligned} &X^{(4)}(x) - \lambda X(x) = 0, , \quad 0 < x < l, \\ &X(0) = X''(0) = 0, \quad X(l) = X''(l) = 0, \end{aligned}$$

其中X''代表二阶导数. 已知常微分方程理论保证此问题的所有本征值$\lambda > 0$ (这一事实可以直接利用，不必证明).

3) (8分) 用分离变量法求解如下问题:

$$\begin{aligned} &\frac{\partial^2 u}{\partial t^2} + a^2 \frac{\partial^4 u}{\partial x^4} = 0, \quad 0 < x < l, \quad t > 0, \\ &u|_{x=0} = \left.\frac{\partial^2 u}{\partial x^2}\right|_{x=0} = 0, \quad u|_{x=l} = \left.\frac{\partial^2 u}{\partial x^2}\right|_{x=l} = 0, \\ &u|_{t=0} = \phi(x), \quad \left.\frac{\partial u}{\partial t}\right|_{t=0} = \psi(x). \end{aligned}$$

附录 在变量代换$x, y \to \xi, \eta$下, 方程

$$a\frac{\partial^2 u}{\partial x^2} + 2b\frac{\partial^2 u}{\partial x \partial y} + c\frac{\partial^2 u}{\partial y^2} + d\frac{\partial u}{\partial x} + e\frac{\partial u}{\partial y} + fu + g = 0$$

变为

$$A\frac{\partial^2 u}{\partial \xi^2} + 2B\frac{\partial^2 u}{\partial \xi \partial \eta} + C\frac{\partial^2 u}{\partial \eta^2} + D\frac{\partial u}{\partial \xi} + E\frac{\partial u}{\partial \eta} + Fu + G = 0,$$

其中

$$\begin{aligned} A &= a(\frac{\partial \xi}{\partial x})^2 + 2b\frac{\partial \xi}{\partial x}\frac{\partial \xi}{\partial y} + c(\frac{\partial \xi}{\partial y})^2, \\ B &= a\frac{\partial \xi}{\partial x}\frac{\partial \eta}{\partial x} + b(\frac{\partial \xi}{\partial x}\frac{\partial \eta}{\partial y} + \frac{\partial \xi}{\partial y}\frac{\partial \eta}{\partial x}) + c\frac{\partial \xi}{\partial y}\frac{\partial \eta}{\partial y}, \\ C &= a(\frac{\partial \eta}{\partial x})^2 + 2b\frac{\partial \eta}{\partial x}\frac{\partial \eta}{\partial y} + c(\frac{\partial \eta}{\partial y})^2, \\ D &= a\frac{\partial^2 \xi}{\partial x^2} + 2b\frac{\partial^2 \xi}{\partial x \partial y} + c\frac{\partial^2 \xi}{\partial y^2} + d\frac{\partial \xi}{\partial x} + e\frac{\partial \xi}{\partial y}, \\ E &= a\frac{\partial^2 \eta}{\partial x^2} + 2b\frac{\partial^2 \eta}{\partial x \partial y} + c\frac{\partial^2 \eta}{\partial y^2} + d\frac{\partial \eta}{\partial x} + e\frac{\partial \eta}{\partial y}, \\ F &= f, \\ G &= g. \end{aligned}$$